

Tutorial 5: Aggregate Demand, Phillips Curve

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Outline

Core Questions

Log-linearization

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- ▶ Does the Phillips curve belong more naturally to the short run, medium run, or long run model of the economy—and how does that choice affect the interpretation of monetary neutrality?
- ▶ Is the NAIRU a “real” structural parameter of the economy or merely a statistical artefact that shifts with institutions, policy, and expectations?
- ▶ If central banks use the Phillips curve as a guide, does that imply an implicit ethical stance about tolerable unemployment and inflation trade-offs?
- ▶ To what extent can the evolution of the Phillips curve—from a stable empirical relationship to the expectations-augmented and New Keynesian versions—be seen as progress in economic science or simply as adaptation to new data?

Log-linearization

Log-linearization: an introduction

- ▶ Way to approximate nonlinear equations around a steady state or eq. point.

Cookbook

- Take logs
 - Do (first order) Taylor approximation around a point of interest (steady state)
 - Simplify so that everything is express in % deviation from steady state
- ▶ Makes equations more tractable for analysis (shocks / policy changes).

Taylor Approximation - 1

- Normally, after taking logs we use Taylor approximation

Taylor Approximation-Univariate

Consider a function $f(x)$ that is differentiable at a point x^* . The first-order Taylor approximation of $f(x)$ around x^* is given by:

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + f''(x^*) \frac{(x - x^*)^2}{2!} + \dots$$

- Normally you do it up to first order $\rightarrow$ linear!!

N.B.: First you take the derivative, then you evaluate it at the point of interest.

Taylor Approximation - 2

Taylor Approximation-Multivariate

Consider a function $f(x, y)$ that is differentiable at a point (x^*, y^*) . The first-order Taylor approximation of $f(x, y)$ around (x^*, y^*) is given by:

$$f(x, y) \approx f(x^*, y^*) + f_x(x^*, y^*)(x - x^*) + f_y(x^*, y^*)(y - y^*) + \dots$$

where $f_x(x^*, y^*) = \frac{\partial f}{\partial x}(x^*, y^*)$ and $f_y(x^*, y^*) = \frac{\partial f}{\partial y}(x^*, y^*)$.

N.B.: First you take the derivative, then you evaluate it at the point of interest.

Exercises

Log-linearize:

- ▶ Cobb-Douglas production function:

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha}$$

- ▶ Accounting identity

$$Y_t = C_t + I_t$$

- ▶ Capital Accumulation Equation

$$K_{t+1} = I_t + (1 - \delta)K_t$$

- ▶ Consumption Euler Equation

$$\left(\frac{C_{t+1}}{C_t} \right)^\sigma = \beta(1 + R_t)$$